

CO2 AT 1

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COURSE: JAVA PROGRAM
COURSE CODE: CSA0924

Q. A returns-processing GUI uses a List component to let staff select several returned items at once for a single bulk refund action, but the existing implementation only supports selecting one item at a time. Identify the AWT setting/mode that enables multi-selection on a List and how the event-handling code retrieving selections must change as a result.

Ans:

Aim: To demonstrate how to enable **multiple selection** in a Java AWT List component and modify the event-handling code to retrieve **all selected returned items** for a single bulk refund action.

Algorithm:

- Start the Java AWT returns-processing application.
- Create a List component containing returned item details.
- Enable multiple selection using `setMultipleMode(true)`.
- Add the List component to the GUI.
- Create a **Bulk Refund** button.
- Add an event listener to the button.
- Check whether any items have been selected.
- If no item is selected, display an appropriate message.
- If items are selected, process each returned item.
- Calculate or display the total refund amount.
- Perform one bulk refund action for all selected items.
- Display the refund details and number of items processed.
- Stop.

Pseudocode:

BEGIN

Create AWT List named returnList

Add returned items to returnList

Enable multiple selection:

 returnList.setMultipleMode(true)

Create "Bulk Refund" button

WHEN button is clicked:

 selectedItems = returnList.getSelectedItems()

 IF selectedItems is empty THEN

 Display "Please select at least one returned item"

 ELSE

 totalRefund = 0

 FOR each item in selectedItems

 Process refund for item

 Add item refund amount to totalRefund

 Display refunded item

 END FOR

 Display total refund amount

 Display number of items refunded

 END IF

END

JAVA CODE:

```
import java.awt.*;
import java.awt.event.*;

public class ReturnsBulkRefundGUI extends Frame implements ActionListener {
    List returnList;
    TextArea output;
    Button refundButton;

    ReturnsBulkRefundGUI() {
        setTitle("Returns Processing - Bulk Refund");
        setSize(700, 500);
        setLayout(new BorderLayout(10, 10));

        Panel topPanel = new Panel(new FlowLayout());
        Label titleLabel = new Label("Select Returned Items for Bulk Refund");
        topPanel.add(titleLabel);

        returnList = new List(8, true);
        returnList.add("RET101 - Keyboard - ₹1200");
        returnList.add("RET102 - Mouse - ₹700");
        returnList.add("RET103 - Monitor - ₹8500");
        returnList.add("RET104 - Headset - ₹1500");
        returnList.add("RET105 - Webcam - ₹2500");
        returnList.add("RET106 - Speaker - ₹3000");
        returnList.add("RET107 - SSD - ₹4500");
        returnList.add("RET108 - RAM - ₹3200");

        returnList.setMultipleMode(true);

        Panel centerPanel = new Panel(new BorderLayout());
        centerPanel.add(returnList, BorderLayout.CENTER);

        Panel buttonPanel = new Panel(new FlowLayout());

        refundButton = new Button("Process Bulk Refund");
        refundButton.addActionListener(this);

        buttonPanel.add(refundButton);

        output = new TextArea(12, 40);
        output.setEditable(false);

        add(topPanel, BorderLayout.NORTH);
        add(centerPanel, BorderLayout.CENTER);
        add(buttonPanel, BorderLayout.SOUTH);
        add(output, BorderLayout.EAST);
    }
}
```

```

addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        dispose();
    }
});

setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    String[] selectedItems = returnList.getSelectedItems();

    if (selectedItems.length == 0) {
        output.setText("Please select at least one returned item.");
        return;
    }

    StringBuilder result = new StringBuilder();

    result.append("BULK REFUND PROCESS\n");
    result.append("-----\n\n");

    double totalRefund = 0;

    for (String item : selectedItems) {
        result.append("Refund Processed: ")
            .append(item)
            .append("\n");

        String[] parts = item.split("₹");

        if (parts.length > 1) {
            try {
                totalRefund += Double.parseDouble(parts[1]);
            } catch (NumberFormatException ex) {
                result.append("Unable to calculate amount.\n");
            }
        }
    }

    result.append("\n-----\n");
    result.append("Total Items Refunded: ")
        .append(selectedItems.length)
        .append("\n");

    result.append("Total Refund Amount: ₹")
        .append(String.format("%.2f", totalRefund))

```

```
        .append("\n");

    result.append("\nBulk refund completed successfully.");

    output.setText(result.toString());
}

public static void main(String[] args) {
    new ReturnsBulkRefundGUI();
}
}
```

OUTPUT

BULK REFUND PROCESS

Refund Processed: RET101 - Keyboard - ₹1200
Refund Processed: RET102 - Mouse - ₹700
Refund Processed: RET104 - Headset - ₹1500

Total Items Refunded: 3
Total Refund Amount: ₹3400.00

Bulk refund completed successfully.